

● EASY

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Lines, angles, and triangles

01

10 answers · Rationale-rich · Teach from doc

Q1**C****C. 40**

Choice C is correct. The angle that is shown as lying between the y° angle and the z° angle is a vertical angle with the x° angle. Since vertical angles are congruent and $x = 65$, the angle between the y° angle and the z° angle measures 65° . Since the 65° angle, the y° angle, and the z° angle are adjacent and form a straight angle, it follows that the sum of the measures of these three angles is 180° , which is represented by the equation $65^\circ + y^\circ + z^\circ = 180^\circ$. It's given that $y = 75$. Substituting 75 for y yields $65^\circ + 75^\circ + z^\circ = 180^\circ$, which can be rewritten as $140^\circ + z^\circ = 180^\circ$. Subtracting 140° from both sides of this equation yields $z^\circ = 40^\circ$. Therefore, $z = 40$.

Choice A is incorrect and may result from finding the value of $x + y$ rather than z . Choices B and D are incorrect and may result from conceptual or computational errors.

Q2**B****B.** 72°

Choice B is correct. It's given that triangle ABC is congruent to triangle DEF . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that angle A corresponds to angle D , and that the measure of angle A is 18° . It's also given that the measures of angles B and E are 90° . Since these angles have equal measure, they are corresponding angles. It follows that angle C corresponds to angle F . Let x° represent the measure of angle C . Since the sum of the measures of the interior angles of a triangle is 180° , it follows that $18^\circ + 90^\circ + x^\circ = 180^\circ$, or $108^\circ + x^\circ = 180^\circ$. Subtracting 108° from both sides of this equation yields $x^\circ = 72^\circ$. Therefore, the measure of angle C is 72° . Since angle C corresponds to angle F , it follows that the measure of angle F is also 72° .

Choice A is incorrect. This is the measure of angle D , not the measure of angle F .

Choice C is incorrect. This is the measure of angle E , not the measure of angle F .

Choice D is incorrect. This is the sum of the measures of angles E and F , not the measure of angle F .

Q3**D****D.** 147

Choice D is correct. It's given that line m is parallel to line n , and line t intersects both lines. It follows that line t is a transversal. When two lines are parallel and intersected by a transversal, exterior angles on the same side of the transversal are supplementary. Thus, $x + 33 = 180$. Subtracting 33 from both sides of this equation yields $x = 147$. Therefore, the value of x is 147.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4

—

Answer not provided in source data — see rationale below.

Choice C is correct. There is a proportional relationship between the height of an object and the length of its shadow. Let c represent the height, in feet, of the cherry tree. The given relationship can be expressed by the proportion $\frac{555}{300} = \frac{c}{16}$. Multiplying both sides of this equation by 16 yields $c = 29.6$. This height is closest to the value given in choice C, 30.

Choices A, B, and D are incorrect and may result from calculation errors.

Q5

B

B. 120

Choice B is correct. The sum of the interior angles of a triangle is 180° . The measures of the two interior angles of the given triangle are shown. Therefore, the measure of the third interior angle is $180^\circ - 70^\circ - 50^\circ = 60^\circ$. The angles of measures x° and 60° are supplementary, so their sum is 180° . Therefore, $x = 180 - 60 = 120$.

Choice A is incorrect and may be the result of misinterpreting x° as supplementary to 70° .

Choice C is incorrect and may be the result of misinterpreting x° as supplementary to 50° .

Choice D is incorrect and may be the result of a calculation error.

Q6**A****A.** 40

Choice A is correct. Given that lines m and n are parallel, the angle marked 130° must be supplementary to the leftmost angle marked a° because they are same-side interior angles. Therefore, $130^\circ + a^\circ = 180^\circ$, which yields $a = 50^\circ$. Lines ℓ and m intersect at a right angle, so lines j , ℓ , and m form a right triangle where the two acute angles are a° and b° . The acute angles of a right triangle are complementary, so $a^\circ + b^\circ = 90^\circ$, which yields $50^\circ + b^\circ = 90^\circ$, and $b = 40$.

Choice B is incorrect. This is the value of a , not b . Choice C is incorrect and may be the result of dividing 130° by 2. Choice D is incorrect and may be the result of multiplying b by 2.

Q7**D****D.** 180

Choice D is correct. In the figure shown, the angle marked x° and the angle marked 142° form a linear pair of angles. If two angles form a linear pair of angles, the sum of the measures of the angles is 180° . Therefore, the value of $x + 142$ is 180.

Choice A is incorrect. This is 90 less than 142, not the sum of x and 142.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the measure, in degrees, of one of the angles shown.

Q8**47**

The correct answer is 47. Based on the figure, the angle with measure x° and the angle with measure 133° together form a straight line. Therefore, these two angles are supplementary, so the sum of their measures is 180° . It follows that $x + 133 = 180$. Subtracting 133 from both sides of this equation yields $x = 47$.

Q9**A****A.** 52

Choice A is correct. The sum of the measures of the three interior angles of a triangle is 180° . Therefore, $31 + 2b + a = 180$. Since it's given that $a = 45$, it follows that $31 + 2b + 45 = 180$, or $2b = 104$. Dividing both sides of this equation by 2 yields $b = 52$.

Choice B is incorrect and may result from a calculation error. Choice C is incorrect. This is the value of $a + 31$. Choice D is incorrect. This is the value of $2b$.

Q10**B****B.** 39

Choice B is correct. The sum of the measures of the interior angles of a triangle is 180 degrees. Since the triangle is a right triangle, it has one angle that measures 90 degrees. Therefore, the sum of the measures, in degrees, of the remaining two angles is $180 - 90$, or 90. It's given that the measure of one of the acute angles in the triangle is 51 degrees. Therefore, the measure, in degrees, of the other acute angle is $90 - 51$, or 39.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the measure, in degrees, of the acute angle whose measure is given.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q4 — the source CB data omitted a structured key for these items, so the answer slot shows —. The worked solution in the rationale below each entry contains the correct value.

Generated 2026-05-29 · cb-educator-question-bank